TRAINING PROGRAMME

# Artificial Intelligence and Big Data Applied to the Healthcare Sector

**Specialty code:** IFCD104

**Official syllabus date:** October 2021

> **Unofficial English translation.** This document translates the official Spanish IFCD104 training syllabus for international review. The Spanish source remains the authoritative version. Layout has been preserved in a restrained document format for readability; this is not a certified legal translation.

## IDENTIFICATION OF THE SPECIALTY AND TRAINING CONTEXT

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|---|---|
| **Specialty name** | ARTIFICIAL INTELLIGENCE AND BIG DATA APPLIED TO THE HEALTHCARE SECTOR |
| **Professional family** | INFORMATION TECHNOLOGY AND COMMUNICATIONS |
| **Professional area** | DEVELOPMENT |
| **Code** | IFCD104 |
| **Professional qualification level** | 3 |

### General objective

Program and apply intelligent systems to optimise management in the healthcare and biosanitary sector.

### Training modules

| Module | Title | Hours |
|---|---|---|
| **1** | Programming, Data and Artificial Intelligence Models | 110 |
| **2** | Machine Learning Systems | 40 |
| **3** | Big Data Systems | 60 |
| **4** | Fundamentals of Probability and Statistics with R | 60 |
| **5** | Technologies Used in Hospitals and Healthcare Systems | 30 |

**Total official specialty duration:** 300 hours.

## DELIVERY MODES AND LEARNER ENTRY REQUIREMENTS

The programme may be delivered in person or through online training. For online delivery, the official syllabus specifies no mandatory in-person tutorial hours.

### Learner access requirements

Applicants must meet at least one of the following:

- Upper-secondary/Bachillerato qualification or equivalent.
- Higher Technician qualification (advanced vocational training) or equivalent.
- Pass the official entry examination for advanced vocational education.
- Pass an official university entrance examination.
- Level-3 professional certificate.
- University degree or equivalent.
- Postgraduate/Master's degree or equivalent.

**Professional experience:** not required.

### Recommended prior knowledge

- Fundamentals of structured programming: basic data types, input/output, control blocks and top-down design.
- Traditional and networked office software.
- English sufficient to read technical documentation, approximately B1 level.

### Trainer requirements

- Relevant university degree or equivalent technical qualification.
- No minimum professional experience is required by the syllabus.
- Documented teaching experience/training or an equivalent recognised trainer qualification.
- For online delivery, trainers must additionally demonstrate training or experience in online teaching and ICT use.

## MINIMUM FACILITIES, EQUIPMENT AND SOFTWARE

| Training space | Minimum surface for 15 participants | Increase per participant |
|---|---|---|
| Management / computer classroom | 45 m² | 2.4 m² per participant (maximum 30 participants) |

### Classroom equipment

- Instructor desk and chair; student desks and chairs; classroom materials and whiteboard.
- Networked computers able to run virtual machines and the software required for each training activity.
- Projector and internet access for the instructor.
- Networked student computers with internet access and printing capability.

### Specified software / platforms

- Ubuntu, Anaconda, Docker.
- Python with NumPy, pandas, Matplotlib and scikit-learn.
- Jupyter Notebooks, TensorFlow, Keras, Colab.
- Hive, Pig, Flume.
- PowerShell.
- Azure, AWS and Google Cloud.
- IaaS, PaaS and SaaS environments.
- R, RStudio and Bioconductor.
- openEHR, Dash and related tooling.

Facilities and equipment must scale with participant numbers, meet applicable industrial and health-and-safety requirements, and provide reasonable accessibility adjustments for learners with disabilities.

### Virtual classroom / online platform requirements

The official syllabus requires synchronous connectivity, bidirectional communication, participant connection records, audio/video, file and screen sharing, and learning platforms capable of supporting collaborative work and tracking participation.

# ONLINE-LEARNING PLATFORM REQUIREMENTS

## Infrastructure

- Capacity and server response appropriate to the enrolled learner population and concurrent usage.
- Bandwidth sufficient for real-time audiovisual communication; the source syllabus specifies a minimum server bandwidth of 300 Mbps.
- 24/7 platform availability.

## Software and interoperability

- Compatibility with SCORM and IMS content packages.
- Web accessibility/interactivity aligned with the applicable Spanish accessibility standard cited in the source syllabus.
- Compliance with applicable Spanish and EU data-protection requirements, including the GDPR framework where relevant.
- Compatibility with common operating systems, databases, browsers and web servers.
- Operational training-tracking web services in accordance with the applicable data model/protocol cited by the source syllabus.

## Services and learner support

- Host the complete virtual learning material for the training specialty.
- Provide centralised technical support throughout the training action.
- Support synchronous and asynchronous communication, including messaging, forums and chat.
- Support group collaboration, user/permission management, content management, assessment, grading and learner-progress tracking.

## Virtual learning materials

Materials must address the programme's objectives and learning outcomes, be pedagogically structured, encourage practical application, use relevant multimedia/interactive resources, support supplementary resources and glossaries, and include formative and final assessment activities.

## RELATED OCCUPATIONS

| Code | Occupation |
|---|---|
| 38201017 | Computer application programmers |
| 27131024 | Web and multimedia analyst-programmers |
| 24211224 | Bioinformaticians / bioinformatics technicians |
| 38111011 | Computer systems operations technicians |
| 27191013 | IT auditors / advisers |

## TRAINING MODULE 1 — PROGRAMMING, DATA AND ARTIFICIAL INTELLIGENCE MODELS

**Duration:** 110 hours

### Objective

Develop applications using Python in a data-analysis context while identifying different artificial intelligence systems.

### Learning outcomes — knowledge, cognitive and practical skills

- **AI foundations:** types of AI, AI applications, AI areas, and the role of data in AI.
- **Python and data tooling:** Anaconda, Jupyter Notebooks, Python and advanced modules including NumPy, pandas, Matplotlib and scikit-learn.
- **Advanced data processing:** structured/unstructured data; acquisition and storage; cleaning; enrichment; standardisation; normalisation; encoding; discretisation; imputation.
- **Data interpretation:** probability distributions, basic statistics, clustering, dimensionality reduction and data visualisation.

## TRAINING MODULE 1 — CONTINUED

### Complex data and engineering practices

- Analysis of tabular data.
- Graph analysis.
- Text-data analysis.
- Time-series analysis.
- Image analysis.
- Good programming and code-review practices.
- Version control systems such as GitHub or equivalent.
- Agile methodologies.
- Technical documentation.

### Management, personal and social skills

- Responsibility in developing, programming and solving basic preprocessing and data-analysis processes.
- Rigour when assessing data quality and predictive capacity.
- Coordination with other team members involved in data-analytics projects.

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## TRAINING MODULE 2 — MACHINE LEARNING SYSTEMS

**Duration:** 40 hours

### Objective

Plan, develop and deploy simple learning systems in a local environment while identifying the main machine-learning approaches.

## TRAINING MODULE 2 — MACHINE LEARNING SYSTEMS

### Learning outcomes

- **Supervised learning foundations:** problem definition, loss functions, learning curves, training/test/validation sets, error analysis and metrics.
- **Classification:** k-nearest neighbours, decision trees, random forests and support-vector machines.
- **Regression:** linear methods, multiple regression, logistic regression and random forests.
- **Neural networks:** TensorFlow development platforms, neural-network regression/classification, computer-vision applications and natural-language-processing applications.
- **Deployment:** machine-learning system lifecycle, containers and APIs using Docker, and cloud services.
- **Limitations and ethics:** appropriate use of learning techniques in decision-making systems, data/bias, uncertainty management and other ethical considerations.

### Management, personal and social skills

- Select an appropriate learning model for a concrete problem.
- Develop and deploy a simple machine-learning application.
- Evaluate machine-learning systems from an ethical perspective.

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## TRAINING MODULE 3 — BIG DATA SYSTEMS

**Duration:** 60 hours

### Objective

Develop and manage simple cloud-based Big Data systems, interpreting their functionality, architecture and components.

## TRAINING MODULE 3 — BIG DATA SYSTEMS

### Learning outcomes

- **Big Data foundations:** Big Data systems, distributed and parallel computing, distributed storage and fault tolerance, and organisational applications.
- **Architectures and tools:** data collection/storage, processing/analysis, and tools/providers such as Hive, Pig and Flume.
- **Common applications:** dashboards, storage-system solutions, processing systems and analytics systems.
- **Cloud practical case:** cloud architecture; IaaS/PaaS/SaaS; Azure, AWS and Google Cloud; PowerShell/scripting; virtual machines; components/architectures; cloud administration; end-to-end simple cloud application development; monitoring, optimisation and troubleshooting; cost planning and management.

### Management, personal and social skills

- Identify when a Big Data system is appropriate and what advantages it offers.
- Critically evaluate Big Data service offerings.
- Develop and put into production a simple cloud Big Data system.
- Administer a cloud Big Data system responsibly.

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## TRAINING MODULE 4 — FUNDAMENTALS OF PROBABILITY AND STATISTICS WITH R

**Duration:** 60 hours

### Objective

Make predictions and analyse data to draw inferences from available information and support decision-making in health sciences.

# TRAINING MODULE 4 — PROBABILITY AND STATISTICS WITH R

## Learning outcomes

- **Scientific knowledge cycle:** scientific method, experimental design, observation/data collection, mathematical/statistical methods, hypotheses and models.
- **Descriptive statistics:** frequency tables and graphical representations including pie charts, bar charts, histograms and frequency polygons.
- **Statistical parameters:** central tendency, dispersion, skewness and shape.
- **Exploratory graphics:** qualitative, discrete quantitative and continuous quantitative data.
- **Probability:** definitions and properties, Laplace's rule, combinatorics and Bayes' theorem.
- **Random variables:** discrete and continuous variables, expected value, variance and standard deviation.
- **Distributions:** binomial and normal distributions.
- **Hypothesis testing:** significance tests/limits, Type I and Type II errors, alpha and beta risks.
- **Biomedical Big Data:** personal medical data, anonymisation, data-protection legislation and biomedical data types.

## Management, personal and social skills

- Identify the origin and type of biomedical-computing data.
- Prepare rigorous reports and visualisations.
- Justify conclusions and predictions from healthcare statistical studies.
- Manage and exploit large healthcare databases responsibly.

## TRAINING MODULE 5 — TECHNOLOGIES USED IN HOSPITALS AND HEALTHCARE SYSTEMS

**Duration:** 30 hours

### Objective

Identify the value chain and business model within the healthcare system.

### Learning outcomes

- **Healthcare-system organisation:** healthcare ecosystem and value chain; resource organisation/planning/location; service-provision models; chronic-care services; technology and health.
- **Management and decision-making:** public-health strategy, health and society, epidemiology and healthcare research.
- **Digital health technologies:** major systems including HIS, LIS, PACS, FIS and EHR.
- **Standards and interoperability:** LOINC, HL7-CDA, HL7-FHIR, SNOMED CT, ICD-10, CPT and openEHR.
- **Clinical documentation:** conceptual framework and core bioinformatics concepts.

### Management, personal and social skills

- Integrate into healthcare-system organisations and clinical/epidemiological research teams.
- Collaborate in the design and programming of AI-project components and tools in the healthcare domain.
- Implement responsible capture, acquisition, manipulation and filtering of health data.
- Apply good practices in data handling in accordance with applicable published standards.

## ASSESSMENT OF LEARNING

- Assessment is theoretical and practical, systematic and continuous throughout each module and at the end of the course.
- An initial diagnostic assessment may be used to establish learners' starting level.
- Methods and instruments must appropriately verify the specified learning outcomes and support reliable, valid assessment.
- Each assessment instrument must include a corresponding correction/scoring system with clear measurement criteria.
- The final outcome is expressed as **Pass / Not Pass**.

## PORTFOLIO CONTEXT

This translation is provided so an international recruiter or technical reviewer can inspect the official scope of the IFCD104 specialty without needing Spanish-language source material.

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|---|---|
| **Official specialty** | IFCD104 |
| **Technical hours** | 300 hours |
| **Portfolio certificate** | 325 total hours including complementary professional training |
| **Core areas** | Python/data/AI · machine learning · Big Data/cloud · probability/statistics with R · healthcare technologies/interoperability |

> The Spanish official syllabus is the authoritative source. This English translation is supplied for readability and professional portfolio review only.